PROBLEM 45

Vibrating Rotor (Unbalanced Turbojet)

Original statement · Problems.pdf, p. 31

Forced Oscillations

To understand the vibrations that occur in a turbojet when the mass distribution of the rotor is slightly unbalanced, we can consider the turbojet as a rigid disk of mass m that rotates at a fixed angular velocity ω, with the center of mass G being separated from the geometric center C by a fixed distance ℓ. An elastic restoring force (with constant k) tries to keep C at its design position O. We call x_0^G, y_0^G the Cartesian coordinates of G, and x_0^C, y_0^C the Cartesian coordinates of C. Considering the weight of the disk negligible, write down the equation of the dynamics of G. Assuming that the homogeneous solution is damped over time and only the particular solution remains, write down the expression of $x_0^G(t)$ and $x_0^C(t)$. Can we affirm that $|\boldsymbol{OC}| > |\boldsymbol{OG}|$ if $\omega > \sqrt{k/m}$?

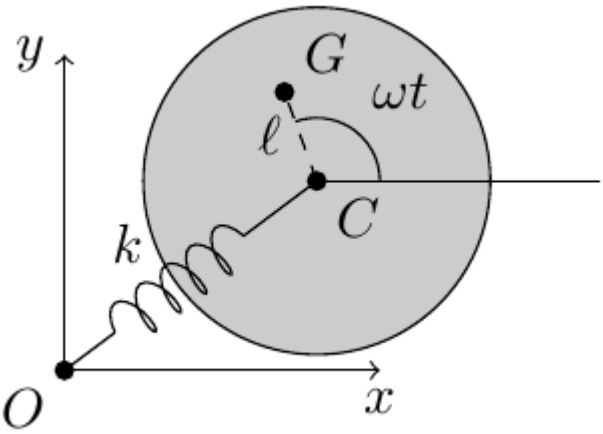


Figure: Rotating disk of centre C with the centre of mass G at distance l, angle omega t from the x direction, spring k from O to C, axes Oxy (Problems.pdf, p. 31).

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** a rigid disk of mass m spins at the constant rate ω. Its geometric centre C is held near the design position O by an elastic restoring force $-k\,\mathbf{r}_C$; its centre of mass G is displaced from C by the fixed distance ℓ (unbalance).
- **Hypotheses:** planar motion in the inertial plane Oxy; the weight is negligible; the spring (zero natural length) acts at C; the only external force is the elastic one, so the motion of G is that of a particle of mass m under that force (Newton's second law applied to G).
- **Parameters and units:** m [kg], k [N/m], ℓ [m], ω [rad/s]; natural frequency $\omega_0 = \sqrt{k/m}$ [rad/s].
- **Unknowns:** $x_0^G(t), y_0^G(t), x_0^C(t), y_0^C(t)$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with $\mathbf{r}_C = x_0^C\,\mathbf{i}_0 + y_0^C\,\mathbf{j}_0$ and $\mathbf{r}_G = x_0^G\,\mathbf{i}_0 + y_0^G\,\mathbf{j}_0$.
- **Unbalance vector:** the segment CG turns with the disk, at the angle ωt from $\mathbf{i}_0$ (figure):

$$\mathbf{r}_G = \mathbf{r}_C + \ell\,\mathbf{e}(t), \qquad \mathbf{e}(t) = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$$

The unit vector $\mathbf{e}$ is the first column of the change-of-basis matrix from the disk-fixed basis (angle ωt about $\mathbf{k}_0$), whose determinant and orthogonality are checked:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

PHASE 3

Step-by-Step Mathematical Deduction

1. Equation of the dynamics of G

Pedagogical Justification: The only external force is the spring force, which depends on the position of C, not on that of G. Newton's second law for the centre of mass, with the position of C eliminated through the geometry $\mathbf{r}_C = \mathbf{r}_G - \ell\,\mathbf{e}(t)$, gives a forced linear equation (Notes Sec. 5.4, oscillator under a periodic force).

$$m\,\ddot{\mathbf{r}}_G = -k\,\mathbf{r}_C = -k\,[\mathbf{r}_G - \ell\,\mathbf{e}(t)]$$

Divide by m and use $\omega_0^2 = k/m$; the two components decouple:

$$\ddot{x}_0^G + \omega_0^2\,x_0^G = \omega_0^2\,\ell\cos\omega t, \qquad \ddot{y}_0^G + \omega_0^2\,y_0^G = \omega_0^2\,\ell\sin\omega t$$

2. Particular solution for G

Pedagogical Justification: The general solution is the homogeneous one, $A\cos\omega_0 t + B\sin\omega_0 t$, which the statement assumes damped out, plus a particular solution. Because the forcing is harmonic at frequency ω, try the same harmonic with an unknown amplitude (method of undetermined coefficients).

$$x_0^G = a\cos\omega t \implies \ddot{x}_0^G = -\omega^2 a\cos\omega t$$

Substitute into the equation of motion:

$$-\omega^2 a + \omega_0^2 a = \omega_0^2\,\ell \implies a = \frac{\omega_0^2\,\ell}{\omega_0^2 - \omega^2} \qquad (\omega \neq \omega_0)$$

The same steps with $y_0^G = b\sin\omega t$ give $b = a$. Therefore

$$x_0^G(t) = \frac{\omega_0^2\,\ell}{\omega_0^2 - \omega^2}\cos\omega t, \qquad y_0^G(t) = \frac{\omega_0^2\,\ell}{\omega_0^2 - \omega^2}\sin\omega t$$

3. Position of the geometric centre C

From $\mathbf{r}_C = \mathbf{r}_G - \ell\,\mathbf{e}(t)$, component by component:

$$x_0^C(t) = \left[\frac{\omega_0^2}{\omega_0^2 - \omega^2} - 1\right]\ell\cos\omega t = \frac{\omega^2\,\ell}{\omega_0^2 - \omega^2}\cos\omega t, \qquad y_0^C(t) = \frac{\omega^2\,\ell}{\omega_0^2 - \omega^2}\sin\omega t$$

Both points describe circles about O at the rotation rate ω. Their distances to O are

$$|\boldsymbol{OG}| = \frac{\omega_0^2\,\ell}{|\omega_0^2 - \omega^2|}, \qquad |\boldsymbol{OC}| = \frac{\omega^2\,\ell}{|\omega_0^2 - \omega^2|}, \qquad \frac{|\boldsymbol{OC}|}{|\boldsymbol{OG}|} = \frac{\omega^2}{\omega_0^2} = \frac{m\,\omega^2}{k}$$

4. Is $|\boldsymbol{OC}| > |\boldsymbol{OG}|$ for $\omega > \sqrt{k/m}$?

Yes. The ratio $|\boldsymbol{OC}|/|\boldsymbol{OG}| = \omega^2/\omega_0^2$ exceeds 1 exactly when $\omega > \omega_0 = \sqrt{k/m}$ (and $\omega \neq \omega_0$, where the particular solution does not exist and the amplitude grows without bound, the resonance). For $\omega < \omega_0$ the opposite holds, $|\boldsymbol{OC}| < |\boldsymbol{OG}|$.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Slow rotation ($\omega \to 0$):** $x_0^G \to \ell$, $x_0^C \to 0$: the disk sits with C at O and G at distance ℓ, as for a static unbalanced disk.
- **Fast rotation ($\omega \gg \omega_0$):** $|\boldsymbol{OG}| \to \ell\,\omega_0^2/\omega^2 \to 0$ and $|\boldsymbol{OC}| \to \ell$. Above the critical speed the disk self-centres: it spins about its centre of mass, and C circles G at the radius ℓ (opposite in phase to the unbalance, because the factor $\omega_0^2 - \omega^2$ is negative).
- **Resonance ($\omega = \omega_0$):** the amplitudes diverge in this undamped model; real rotors cross this critical speed quickly or are damped.
- **Dimensional check:** $[\omega_0^2\ell/(\omega_0^2 - \omega^2)] = \mathrm{m}$ and $[k/m] = \mathrm{N/m/kg} = \mathrm{s}^{-2} = [\omega^2]$.

- **Symbolic check (sympy):** the two particular solutions satisfy the equations of motion with zero residual, and $x_0^G - \ell \cos \omega t$ equals $\omega^2 \ell \cos \omega t / (\omega_0^2 - \omega^2)$.

Source: Problems.pdf